

Roll No.

TCS-726

**B. TECH. (CST) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

ARTIFICIAL INTELLIGENCE AND PATTERN RECOGNITION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub-questions among (a) and (b) in each main question.

(iii) Each question carries 10 marks.

1. (a) Explain intelligence and artificial intelligence. How do you distinguish between the two ? Write a short note on Turing Test. (CO1)

(b) What are state space search and problem formulation ? Discuss the relative merits of Depth-first and Breadth-first searching methods.

(CO3)

2. (a) Explain the concept of 'The Natural Language Processing System'. Describe in detail the difference between natural language understanding and language generation. Explain transition Net, RTN and ATN in NLP. (CO2)

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(2)

(b) "All connectionist models have symbolic components and all symbolic models have mathematical mechanisms." State whether you support or oppose the statement. Justify your response. (CO1)

3. (a) How to specify a task environment, illustrating the process with an example. What do you mean by PEAS ? Determine the PEAS values for the following AI problems : (CO1)

Taxi Driver.

(b) Explain AI agents design philosophy using neat and clean diagrams. (CO2)

4. (a) Explain the properties of Environment in designing intelligent agents. (CO3)

(b) What is the difference between Strong Artificial Intelligence and Weak Artificial Intelligence ? Write short notes on NLP, ML and DL. (CO1)

5. (a) Dehradun city can be represented as a graph by taking landmarks (such as Clock Tower, Railway station, ISBT, GEU etc.) as nodes and the roads as the edges that connect the nodes in the graph. Give a solution to generate the shortest route for Graphic Era bus routes which can be used to give directions for real time navigation using search algorithms. Justify your response. (CO2)

(b) What are the different types of Intelligent agents ? Explain simple reflex and model based intelligent agent with suitable examples. (CO3)